

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Tuesday 20 May 2025

Afternoon (Time: 40 minutes)

Paper reference **31617H/1B**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and functions of cells and tissues (Biology)
 - Section B: Periodicity and properties of elements (Chemistry)
 - Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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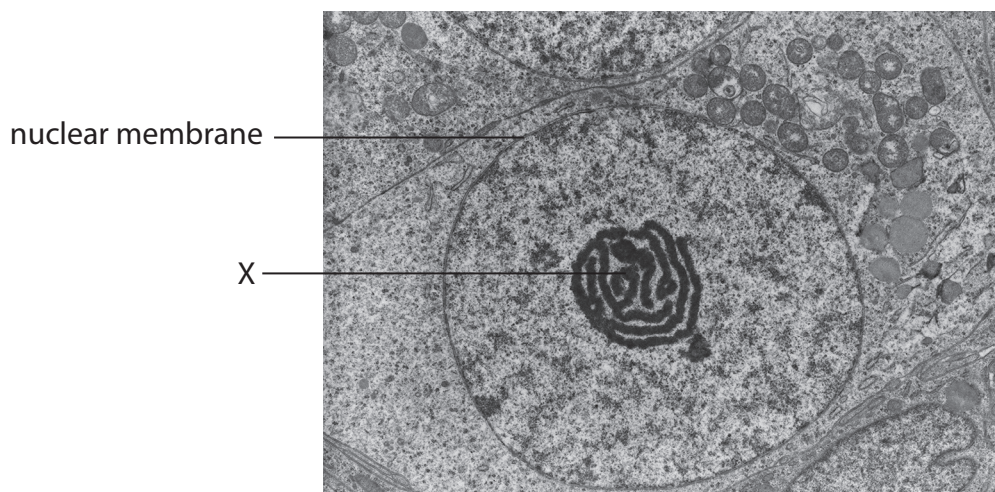

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1** Animal cells and plant cells are eukaryotic cells.

Figure 1 shows an electron micrograph of a cross section of a nucleus.



(Source: © PAL)

Figure 1

- (a) (i) Name the structure labelled X in Figure 1.

(1)

- (ii) Complete Table 1, using ticks to show which organelles are found in animal cells and which organelles are found in plant cells.

(3)

The first row has been completed.

organelles	found in an animal cell	found in a plant cell
Golgi apparatus	✓	✓
chloroplasts		
plasmodesmata		
ribosomes		

Table 1



Microscopes magnify the size of actual objects so that they can be viewed as an image in greater detail.

(b) Which equation is used to calculate magnification?

(1)

- ☐ **A** $\frac{\text{size of actual object}}{\text{size of image}}$
- ☐ **B** size of actual object $\times$ size of image
- ☐ **C** $\frac{\text{size of image}}{\text{size of actual object}}$
- ☐ **D** size of image + size of actual object

(Total for Question 1 = 5 marks)



2 Doctors use electrocardiograms (ECGs) to diagnose cardiovascular diseases.

Figure 2 shows an ECG trace.

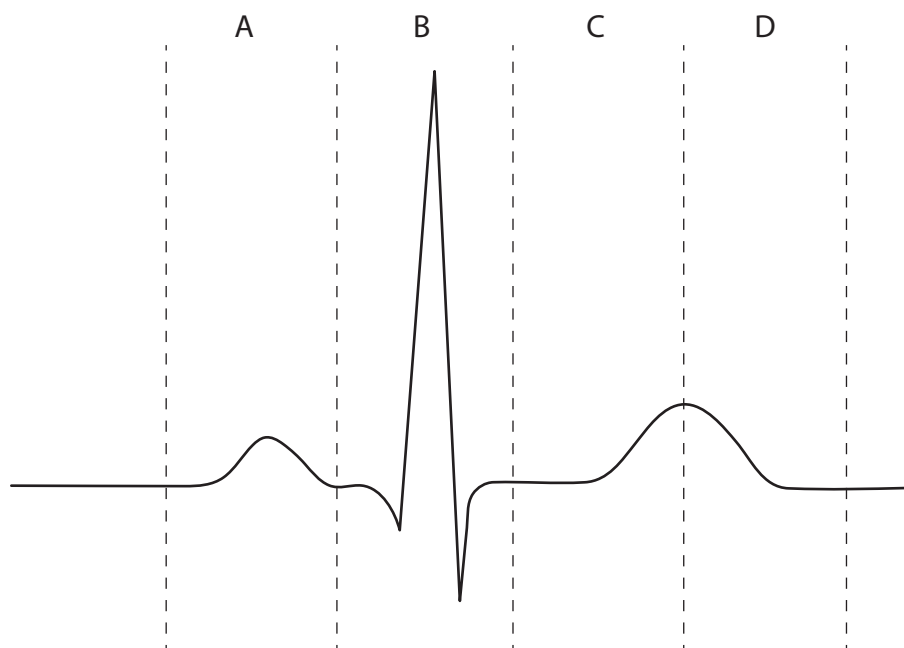


Figure 2

(a) Identify the part of the ECG trace that indicates the contraction of the ventricles.

(1)

- ☐ **A** part A
- ☐ **B** part B
- ☐ **C** part C
- ☐ **D** part D

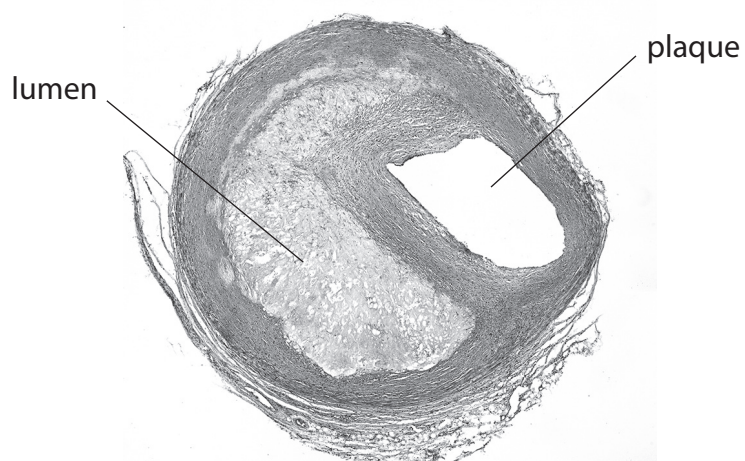
Atherosclerosis is the build-up of plaque in the arteries and can be caused by an unhealthy diet.

(b) (i) State **one** component in a person's diet that can cause plaque to form.

(1)



Figure 3 shows the lumen of an artery with a plaque.



(Source: © PAL)

Figure 3

The lumen of the artery originally had a diameter of 4.2 mm.

The artery developed a plaque.

The lumen of the artery then had a diameter of 2.9 mm.

(ii) Calculate the percentage decrease in the diameter of the lumen.

(3)

Show your working.

percentage decrease in the diameter of the lumen = %



(iii) Explain how white blood cells are involved in the formation of plaques.

(3)

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(Total for Question 2 = 8 marks)

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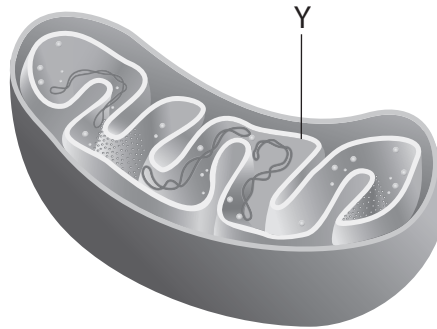
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QUESTION 3 STARTS ON THE NEXT PAGE



P 8 0 4 7 8 A 0 7 1 6

- 3 Muscle cells have more mitochondria than most other cell types.

Figure 4 shows a diagram of a mitochondrion.



(Source: © PAL)

Figure 4

- (a) Name structure Y in Figure 4.

(1)

The mitochondria in muscle fibres produce ATP.

ATP is used as the energy source for myofibril contraction.

- (b) Describe the role of ATP in myofibril contraction.

(2)



People with a higher percentage of slow-twitch muscle fibres are better at running marathons than sprinting short distances.

(c) Explain why slow-twitch muscle fibres need a dense network of capillaries.

(3)

(Total for Question 3 = 6 marks)



4 White blood cells protect the body from pathogens.

Phagocytes are one type of white blood cell.

Phagocytes can digest pathogens through a process called phagocytosis.

Figure 5 shows some of the steps involved in phagocytosis.

(a) Complete Figure 5.

(1)

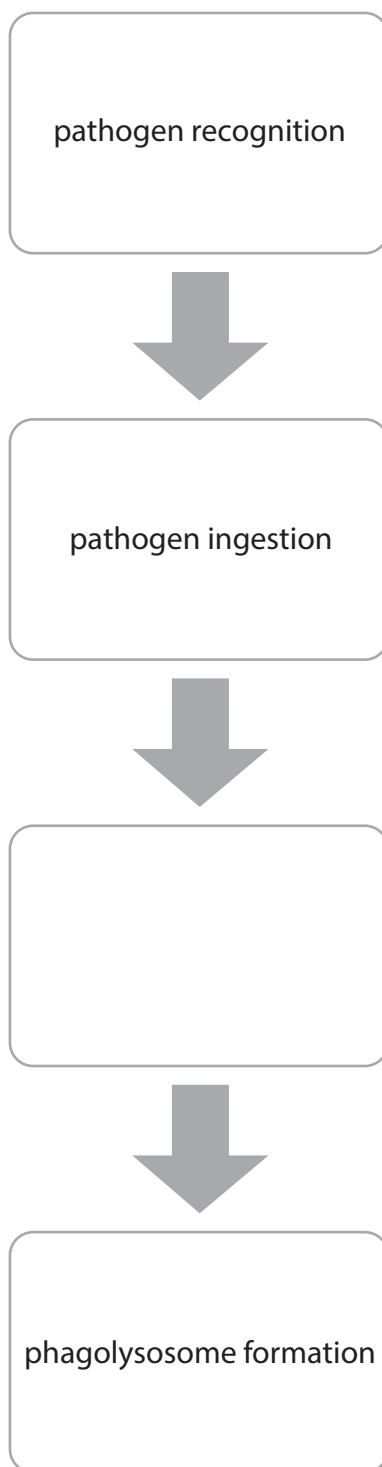


Figure 5

Lymphocytes are another type of white blood cell.

Lymphocytes produce antibodies.

Figure 6 shows the antibody concentration in blood after first exposure to a pathogen and after second exposure to the same pathogen.

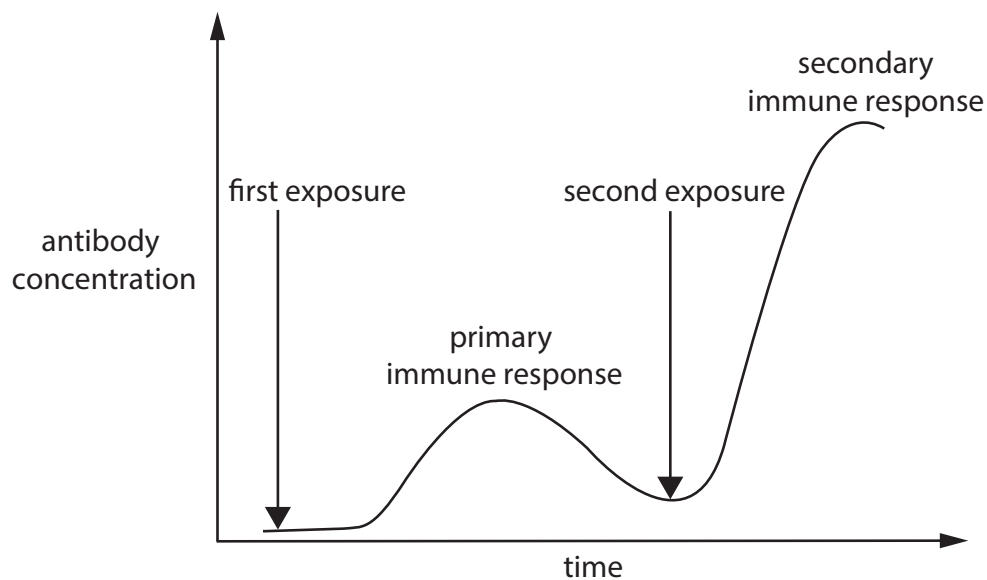


Figure 6

(b) Explain the differences in antibody concentration after first and second exposure.

(4)

Use information in Figure 6 to support your answer.

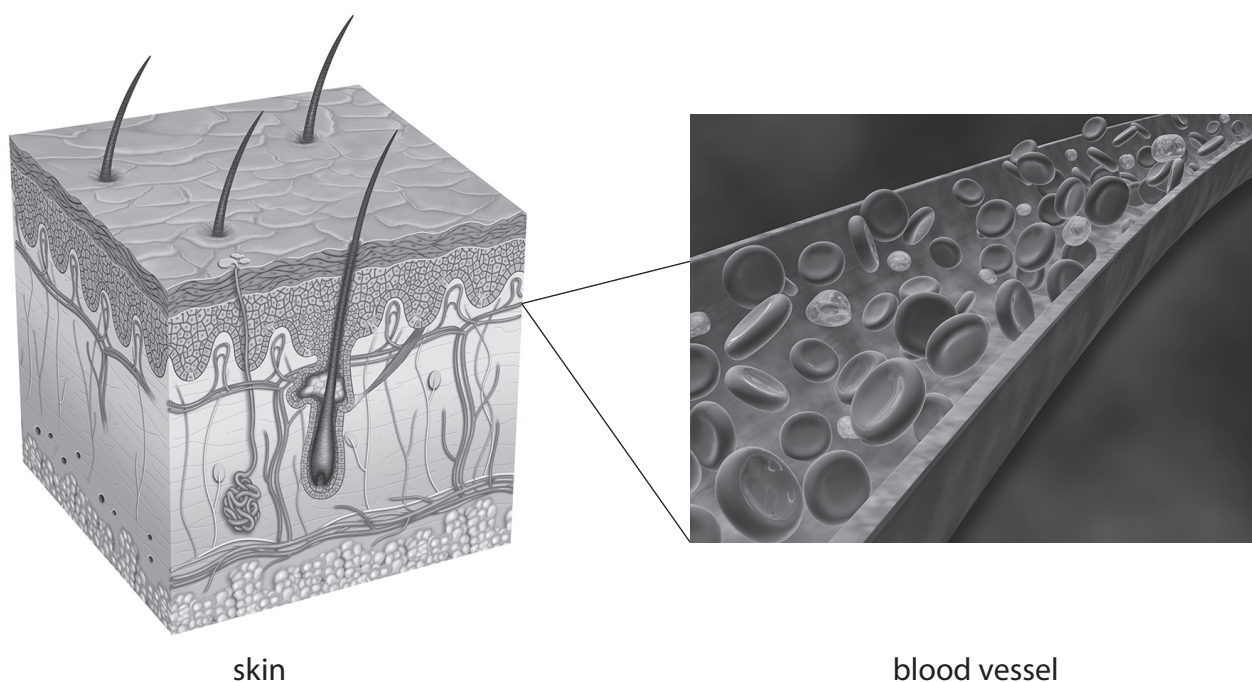
(Total for Question 4 = 5 marks)



5 Multicellular organisms have different levels of organisation.

The levels of organisation can be defined by cell theory.

Figure 7 shows cross sections through the skin and a blood vessel in the skin.



(Source: © PAL)

Figure 7

Explain the levels of organisation for skin **and** for blood in the blood vessel.

(6)

Use cell theory and Figure 7 to support your answer.

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TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



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